

Does Spatial Correlation of Carbon Intensity Improve Robust Carbon-Aware Data-Center Scheduling?

A Multi-Grid Falsification Study: Causal Mechanism, Copula Test, and an Active-Transfer Extension

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Background & research questions

Data centers consume 1–2% of global electricity, and compute is unusually **flexible**, batch/training jobs can shift in time and across sites. Carbon intensity varies strongly by hour and region, so a scheduler can defer work to cleaner moments. Robust scheduling under *uncertain* carbon intensity uses **distributionally robust optimization (DRO)**. A natural extension treats carbon intensity as a stochastic *vector* so that **spatial correlation** can inform the schedule.

- RQ1.** Is carbon intensity of coupled grids genuinely correlated?
RQ2. Does feeding that correlation to a covariance-based DRO reduce out-of-sample tail emissions (CVaR_{0.95})?
RQ3. If not, *why*, and what dependence structure would a richer model need?

Method: a falsification test

A Mahalanobis–Wasserstein DRO, solved as a second-order cone program:

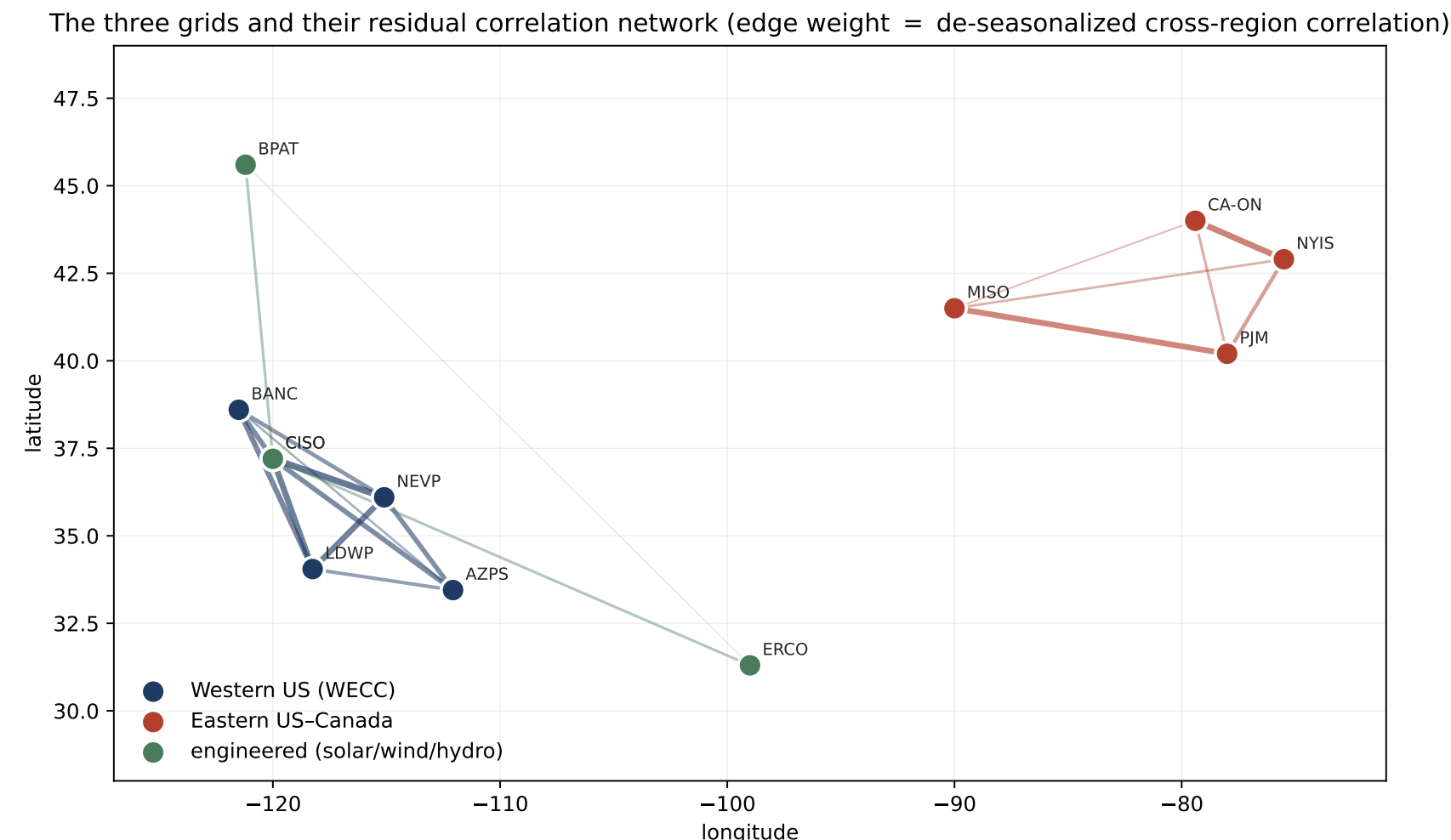
$$\min_{x \in X, t \geq 0} \langle \bar{\rho}, x \rangle + \varepsilon t \quad \text{s.t.} \quad \|L^\top \text{vec}(x)\|_2 \leq t, \quad LL^\top = \hat{\Sigma}.$$

Shuffled-marginals test. Fit the schedule twice, on the *joint* covariance vs. a *block-diagonal* one with all cross-region structure zeroed, and compare out-of-sample CVaR_{0.95} of daily emissions. The *spatial gap* is positive iff the joint covariance helps.

- Pre-registered: config commit-locked, single 2025 test read.
- Five region sets: 3 US/Canada grids spanning low-to-strong correlation ($\bar{r} \approx 0.2$ to 0.65) + a near-zero Iberia–France anchor.
- Blocked 5-fold CV selects ε^* ; 1000-bootstrap CIs.

RQ1: the correlation is real

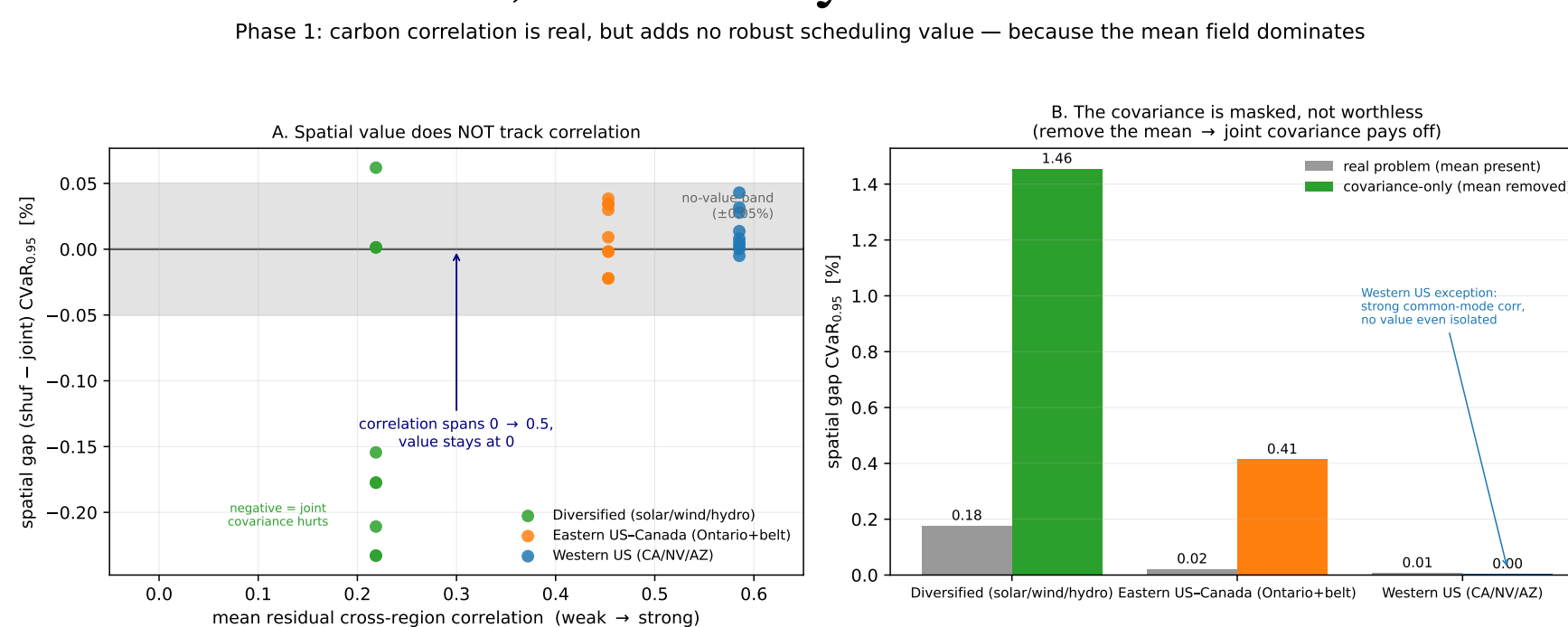
After removing each zone’s hour-of-day mean, residual cross-region correlation is **strong and survives de-seasonalization**: US West CISO–NEVP 0.78; Ontario belt CA-ON–NYISO 0.73. The three grids and their correlation network:



The assumption motivating spatial DRO is empirically valid.

RQ2, but it adds no robust value

Across the dependence spectrum the spatial gap **never exceeds a few tenths of one percent** and is not sign-stable. It survives Ledoit–Wolf shrinkage, seasonal/AR(1) residualization, Benjamini–Hochberg correction over 144 cells, walk-forward to 2024, tighter-ramp and utilization sweeps, and the full tail range CVaR_{0.90}–CVaR_{0.99}. A per-cell **equivalence test (TOST)** affirmatively rejects any effect above the 0.4% materiality margin in all 27 cells, the gap is statistically = 0, not merely undetected.

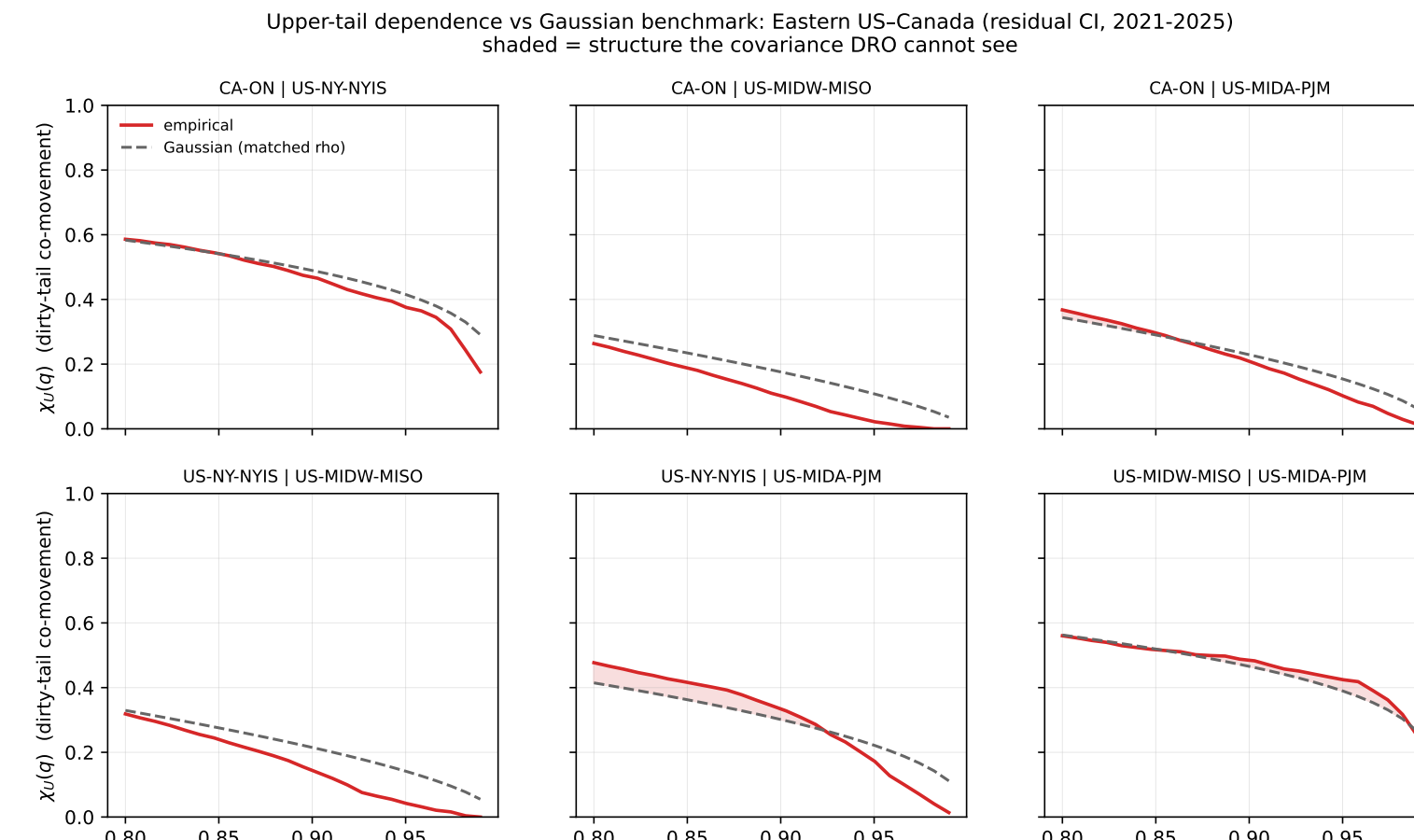


This is an **active null**: CV selects $\varepsilon^* = 1$ in nearly every cell, so the DRO is genuinely robustifying, and the covariance *still* buys nothing.

RQ3, the mechanism

Mean-ablation (causal). Flatten the mean carbon field and the joint covariance suddenly pays, up to +1.46%. So the spatial signal is **real but masked** by the diurnal mean field, not absent.

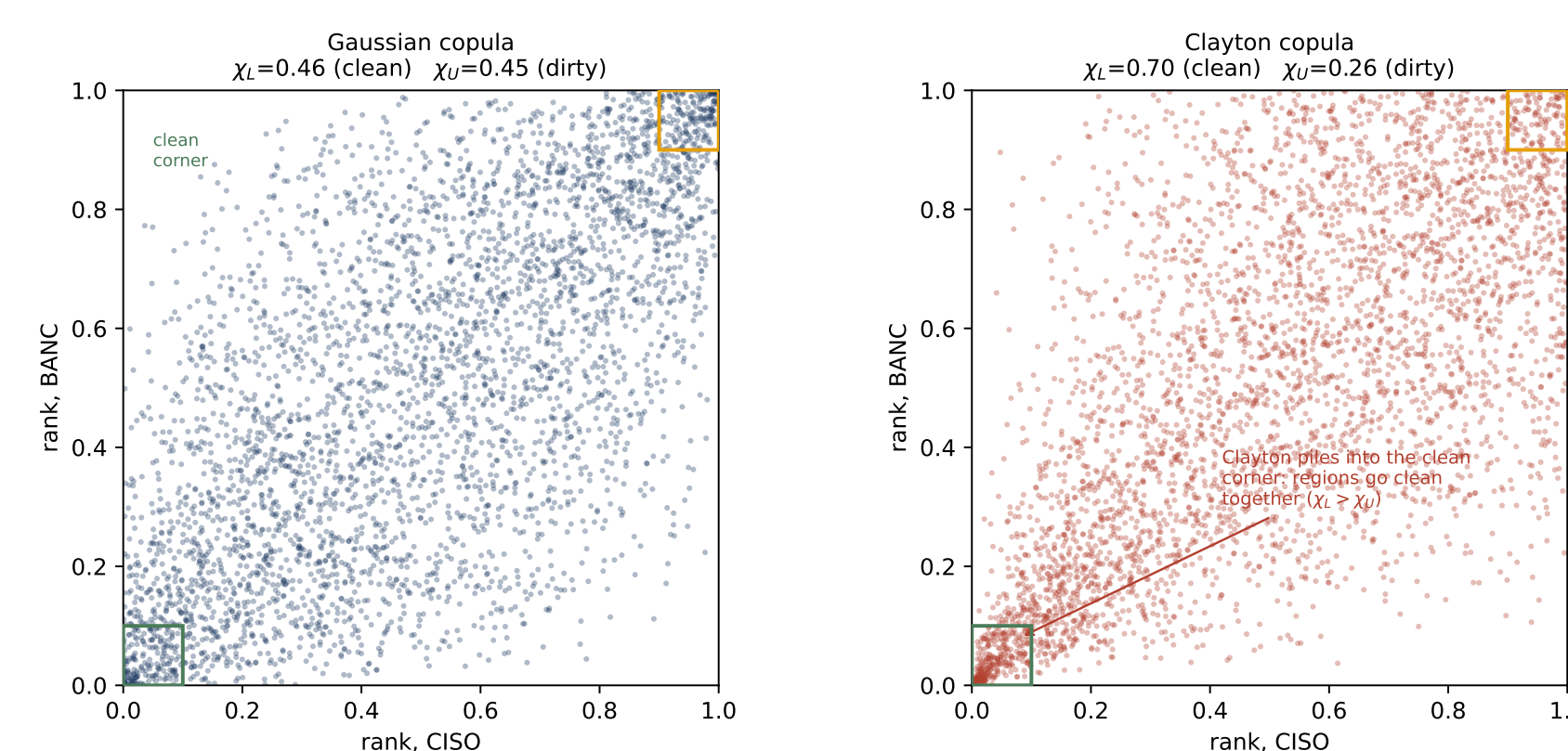
Tail dependence (structural). The residual dependence is **non-elliptical**: regions go *clean* together more than *dirty* together ($\chi_L > \chi_U$, upper-tail independent). An elliptical covariance ball forces $\chi_L = \chi_U$, it is the *wrong object*.



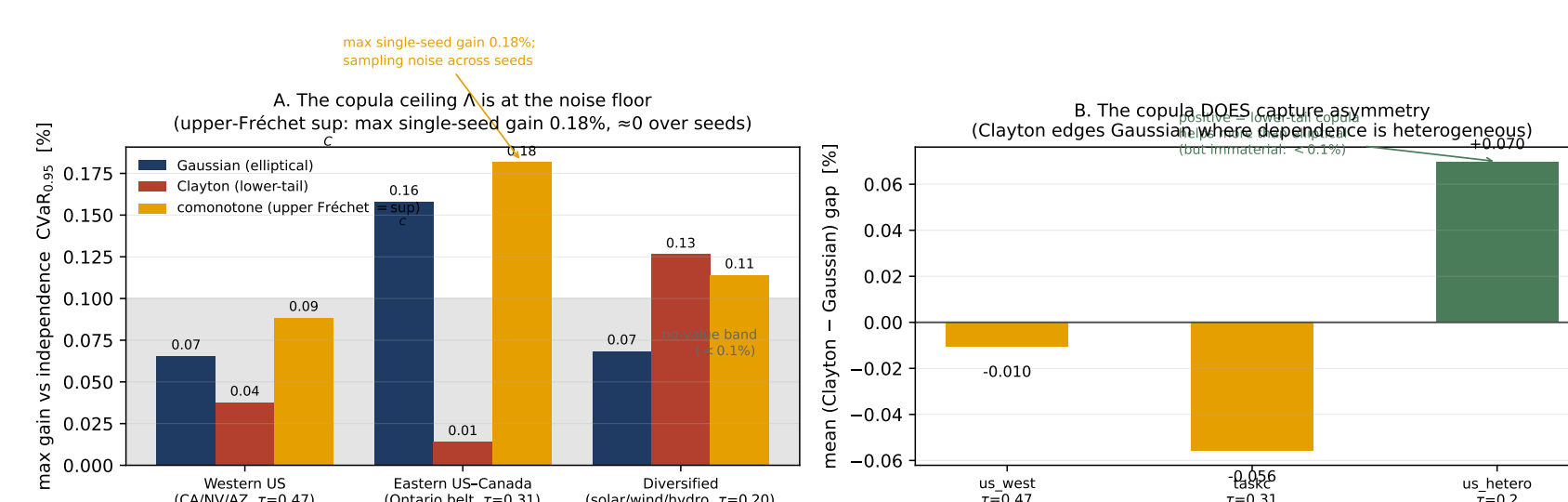
Phase 2, even the right copula fails

We replace the covariance ball with copula schedulers (CVaR sample-average) that *can* represent the asymmetry: independence, Gaussian, lower-tail **Clayton**, and the maximal **comonotone** (upper Fréchet sup_C).

The dependence object a covariance ball cannot represent: elliptical (symmetric) vs. lower-tail copula



Phase 2: even a copula built for the $x_i > x_j$ structure recovers no material scheduling value — the mean field dominates



Even the *maximal* copula’s gain sits at the scenario-noise floor ($|\Lambda| \lesssim 0.1\%$, ≈ 0 over seeds): the null is **total**.

A mean-dominance bound

By translation invariance of CVaR, $\langle \rho, x \rangle = \langle \bar{\rho}, x \rangle + \langle \xi, x \rangle$, the dependence model moves only the residual term.

Proposition. For the SOCP, the spatial gap obeys $|\text{OPT}(\Sigma) - \text{OPT}(\Sigma^{\text{shuf}})| \leq \varepsilon (\max_x \|x\|_2) \|\Sigma^{\text{off}}\|_2^{1/2}$, vanishing as $\varepsilon \rightarrow 0$ or $\Sigma^{\text{off}} \rightarrow 0$.

The tight ceiling is *measured* by the comonotone arm ($|\Lambda| \lesssim 0.1\%$, at the scenario-noise floor), an order below the mean-driven value.

Conclusions & the active extension

- Spatial correlation of carbon is **real but adds no robust scheduling value**, anywhere on the spectrum, for *any* passive dependence model.
- The binding constraint is the **mean field**, not the shape of the dependence (mean-dominance bound + comonotone ceiling).
- **Recommendation:** per-region marginal schedulers capture essentially all the value at a fraction of the cost.
- **We build the one route the bound leaves open**, an *active* inter-region transfer channel $f_{r \rightarrow s, t}$. It cuts **4–10% deterministically** (by exploiting the spatial mean), but *robustifying* it pays nowhere data-grounded, online or under real tail-day emergencies. The null extends to the active and online settings: value lives in the decision, not the dependence model.

Reproducible pipeline: **187 unit tests**, pre-registered configs, per-cell equivalence tests, multi-seed stability, an independent adversarial review, license-safe archived results. *AI used as a coding/drafting assistant under the author’s direction.*